The Liouville skeleton in the skew-symmetrizable acyclic case

Definitions, theorem, and verified Lean correspondence

Cluster skeleton formalization project

Abstract

We state the skeleton theorem for an integral matrix E with determinant one and an arbitrary positive rational diagonal matrix $\widetilde{D}$ such that $K = E\widetilde{D}$ is skew-symmetric. The Lean formalization constructs the real smooth Liouville geometry and proves the full dynamical skeleton identity for all these inputs. The accompanying blueprint gives the weighted geometric argument and explains the scope of the formal result.

1 Input data and exchange space

Let $n \geq 0$, let $M \subseteq \{1, \dots, n\}$ be mutable, and let F be its complement. Fix

$$E \in M_n(\mathbb{Z}), \quad \det E = 1, \qquad \widetilde{D} = \operatorname{diag}(d_1, \dots, d_n), \quad d_j \in \mathbb{Q}_{>0}, \qquad K = E\widetilde{D}, \quad K^{\mathsf{T}} = -K.$$

Thus $E_{ij}d_j = -E_{ji}d_i$, rather than $E_{ij} = -E_{ji}$. Require the mutable block $B = E_{M,M}$ to be acyclic: the arrows $i \to j$ with $B_{ij} > 0$ have no directed cycle. No acyclicity is required on the full set of indices. The unimodular completion is part of the input; existence of such a completion for every exchange matrix is not asserted. Equivalently one may require $E \in GL_n(\mathbb{Z})$: a nonsingular real skew-symmetric matrix has positive determinant, so $\det E = \det K / \det \widetilde{D} > 0$, while unimodularity gives $\det E = \pm 1$.

Use *column* exchange exponents throughout:

$$P_i^{\pm}(z) = \prod_{j=1}^{n} z_j^{[\pm E_{ji}]_+}.$$

$$X_E = \left\{ (z, z') \in \mathbb{C}^n \times \mathbb{C}^M : \begin{array}{ll} z_i z_i' = P_i^+(z) + P_i^-(z) & (i \in M), \\ z_f \neq 0 & (f \in F) \end{array} \right\}.$$

Here $[a]_+ = \max(a, 0)$. Give X_E the induced topology. Frozen coordinates remain variable elements of $\mathbb{C}^*$; no principal-coefficient assumption is made. The initial torus is $T = \{z_1 \cdots z_n \neq 0\} \subset X_E$. On it set

$$c_i = Ee_i, \qquad t_i = z^{c_i} = P_i^+ / P_i^-, \qquad \Omega_T = \frac{1}{2} \sum_{j,l} K_{jl} \, \mathrm{d}\log z_j \wedge \mathrm{d}\log z_l.$$

The integer exponents come from E, not from the generally rational K.

2 Independently defined subsets

Define the coordinate cap without reference to any flow:

$$C_E = \{(z, z') \in X_E : |z_f| = 1 \ (f \in F), \quad |z_i| \le 1, \ (1 - |z_i|^2) z'_i = 0 \ (i \in M)\}.$$

Call $I \subseteq M$ independent if $E_{ij} = 0$ for all $i, j \in I$, and put

$$\Lambda_I = \{(z, z') \in X_E : z'_i = 0, \ |z_i| \le 1 \ (i \in I), \quad |z_j| = 1 \ (j \notin I)\}.$$

For a real flow Φ on X_E, independently define

$$\mathrm{Skel}(\Phi) = \{p \in X_E : \overline{\{\Phi_t(p) : t \ge 0\}}^{X_E} \text{ is compact}\}.$$

The closure is taken in the actual exchange space.

3 The weighted geometric theorem

Theorem 3.1 (Skew-symmetrizable skeleton theorem)**.** *For every $(n, E, \widetilde{D}, M)$ satisfying the hypotheses above, the fixed space X_E admits a smooth real manifold structure of dimension $2n$, a smooth real one-form λ, a smooth vector field Z, and a jointly smooth flow $\Phi : \mathbb{R} \times X_E \to X_E$ such that:*

1. *The actual coordinate inclusion in $\mathbb{C}^n \times \mathbb{C}^M$ is a smooth immersion, and T is dense.*
2. *$\omega = \mathrm{d}\lambda$ is nondegenerate and $\omega|_T = \operatorname{Im} \Omega_T$.*
3. *$\iota_Z \omega = \lambda$, $\Phi_0(p) = p$, $\Phi_{t+s}(p) = \Phi_t(\Phi_s(p))$, and $\frac{\mathrm{d}}{\mathrm{d}t}\Phi_t(p) = Z(\Phi_t(p))$ for every real s, t. Thus this same field is complete in both time directions.*
4. *For this same flow,*

$$\boxed{\mathrm{Skel}(\Phi) = C_E = \bigcup_{I \subseteq M \text{ independent}} \Lambda_I.}$$

   Each Λ_I in this union is compact.

For independent I, the node chart $U_I = \{z_j \ne 0 \ (j \notin I)\}$ has the form $\mathcal{N}^I \times (\mathbb{C}^*)^{n-2|I|}$, where $\mathcal{N} = \{(x, y, t) : xy = 1 + t, \ t \ne 0\}$; in particular $2|I| \le n$.

The quantifier order is $\forall (n, E, \widetilde{D}, M) \ \exists$(smooth structure, λ, Z, Φ). No flow, primitive, completeness, nondegeneracy, or skeleton inclusion is supplied as an additional hypothesis. The real smooth theorem is proved in Lean; the blueprint also explains the local extension of Ω_T.

4 Why the weights cannot be discarded

Write $z_j = e^{u_j + \mathrm{i}\theta_j}$, $v_i = c_i(u)$ and $\eta_i = c_i(\theta)$. The blueprint constructs

$$\lambda_{\mathrm{tor}} = u^{\mathsf{T}} K \, \mathrm{d}\theta, \qquad H = \sum_{i \in M} d_i h(u_i) \chi_i \rho(v_i) \beta(\eta_i), \qquad \lambda = \lambda_{\mathrm{tor}} + \mathrm{d}H.$$

The identities $K e_i = d_i c_i$ and $c_i^{\mathsf{T}} K^{-1} = -e_i^{\mathsf{T}}/d_i$ give

$$\dot{u}_i = u_i - h(u_i) \chi_i \rho(v_i) \beta'(\eta_i), \qquad \dot{\eta}_i = d_i^{-1} \partial_{u_i} H.$$

The radial equations have the same form as in the unweighted case because the weights cancel, not because d_i is assumed to be one. Every node symplectic block and its primitive carries the factor d_i. Positivity of d_i and invertibility of K enter the construction.

A concrete non-skew input covered by the universal theorem is

$$E = \begin{pmatrix} 0 & 1 & -1 & 0 \\ -2 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \widetilde{D} = \operatorname{diag}(1, 2, 1, 2), \qquad M = \{1, 2\}.$$

Here $\det E = 1$, $E\widetilde{D}$ is skew-symmetric, and the mutable quiver is acyclic, but $E_{12} = 1 \neq 2 = -E_{21}$. This example checks that the input is genuinely broader; it does not replace the theorem quantified over all admissible inputs.

5 Lean correspondence and verification

`Seed` contains the integral E, positive rational weights d_i, weighted skew-symmetry, mutable indices and mutable acyclicity. `Seed.Unimodular` means $\det E = 1$. `FullSkeleton.firstGeometricTarget` proves the universally quantified existence theorem. The exchange monomials use column exponents definitionally, and the torus form uses $K = E\widetilde{D}$ with the sign and factor $1/2$ displayed above.

The library proves both the cap and stratified skeleton identities, constructs the node charts and smooth structure, and proves the chart rank and dimension bound. `PdfTargetConsumer.pdf_target`, in `verification/FullSkeletonConsumer.lean`, assembles the published clauses into one checked statement. The repository's proof-to-code map lists the individual declarations.

The $\widetilde{D} = I$ specialization is recovered by `SkewSeed.skeleton_recovery`. On the old exchange-point carrier it gives the original torus and cap descriptions and the skeleton of the conjugate flow at the same times. The identification of carriers is the identity on ambient coordinates; the logarithmic form uses the old matrix B with no sign change. Smooth witnesses are existential on the weighted reading of that space. No term identity with the former library's particular atlas, primitive or flow is asserted.

The displayed non-skew example and its side conditions are checked in `Examples`; its geometry is an instance of the universal theorem.

The committed verification receipt checks 65 modules, 1811 declarations, 72 named audit roots and 73 hashed inputs, including the final consumers. The only permitted transitive axioms are `propext`, `Classical.choice`, and `Quot.sound`. The pinned dependencies are unmodified. The exact source hashes, logs and statement review are in `verification/`.

6 Scope

The formal input uses $\det E = 1$ explicitly. The elementary equivalence with an integral unimodular completion under the weighted skewness hypothesis is explained in Section 1; the determinant-positivity argument itself is not a separate theorem in this library. The result concerns real smooth Liouville geometry. It does not claim a separately formalized holomorphic extension of Ω_T, singular completions, arbitrary cyclic seeds, cap-product descriptions, generalized cocores, or categorical mirror symmetry. Frozen coordinates remain variable and no principal-coefficient restriction is imposed.